Kimberly Navarro

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Q1.)

Protein name: Insulin receptor isoform B preproprotein

Accession Number: NP_001316985.1

Species: Mus musculus

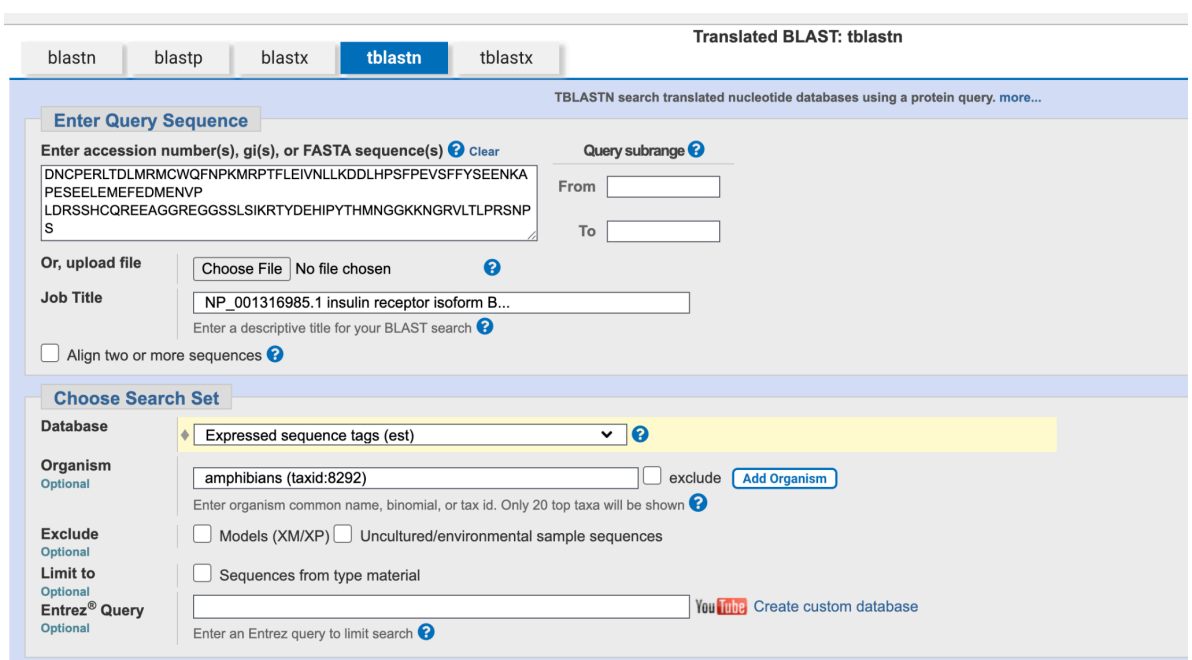
Function: This gene encodes a member of the receptor tyrosine kinase family of transmembrane signaling proteins that play important roles in cell differentiation, growth and metabolism.

Q2.)

Method: TBLASTN search searched against amphibians (taxid:8292) ESTs

Database: Expressed sequence tags (est)

Organism: amphibians (taxid:8292)



BLAST Results:

Your search is limited to records **include: amphibians (taxid:8292)**

Job Title	**NP_001316985.1 insulin receptor isoform B...**
RID	RZGE3JSN016 *Search expires on 02-03 04:12 am* Download All
Program	TBLASTN Citation
Database	est See details
Query ID	lcl|Query_6556701
Description	NP_001316985.1 insulin receptor isoform B preproprotein ...
Molecule type	amino acid
Query Length	1384
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Chosen match:

Accession: BU913403.1, from Xenopus laevis

E-value: 1e-136, Perc identity: 71.84%, Length: 990

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
CBXT21134.b1 NICHD_XGC_trop_25 Xenopus tropicalis cDNA clone IMAGE:9018941 5', mRNA sequence	Xenopus tropicalis	476	476	19%	1e-156	82.40%	809	EL867388.1
AGENCOURT_10495233 NICHD_XGC_OO1 Xenopus laevis cDNA clone IMAGE:6639745 5', mRNA sequence	Xenopus laevis	426	426	21%	1e-136	71.48%	990	BU913403.1
BX852271 Wellcome CRC pRN3 dorsal lip Xenopus laevis cDNA clone IMAGp998K148489; IMAGE:3473797 5', ...	Xenopus laevis	410	410	16%	3e-132	86.82%	676	BX852271.1
de34f07.y1 Wellcome CRC pRN3 dorsal lip Xenopus laevis cDNA clone IMAGE:3473797 5' similar to TR:Q9PVZ4 ...	Xenopus laevis	406	406	17%	1e-130	82.33%	700	BG162445.1
AGENCOURT_54839250 NICHD_XGC_Emb10 Xenopus laevis cDNA clone IMAGE:7982235 5', mRNA sequence	Xenopus laevis	395	395	17%	2e-126	78.90%	758	DR722802.1
AGENCOURT_15183035 NICHD_XGC_Emb4 Xenopus laevis cDNA clone IMAGE:5514314 5', mRNA sequence	Xenopus laevis	387	387	17%	2e-123	77.12%	770	CF287002.1
AGENCOURT_10485061 NICHD_XGC_Emb1 Xenopus laevis cDNA clone IMAGE:6637443 5', mRNA sequence	Xenopus laevis	369	369	19%	2e-115	67.30%	954	BU911460.1

Descriptions | Graphic Summary | **Alignments** | Taxonomy

Alignment view Pairwise Restore defaults

3 sequences selected

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AGENCOURT_10495233 NICHD_XGC_OO1 Xenopus laevis cDNA clone IMAGE:6639745 5', mRNA sequence

Sequence ID: BU913403.1 **Length:** 990 **Number of Matches:** 1

Range 1: 58 to 915 GenBank Graphics Next Match Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
426 bits(1095)	1e-136	Compositional matrix adjust.	208/291(71%)	235/291(80%)	6/291(2%)	+1

```
Query  961   IGPLIFVFLFSVVIGSIYLFLRKRQPDG-PMGPLYASSNPEYLSASDVFPSSVYVPDEWE  1019
             I P+I +   +++   I+ + +K   +G P G LYAS NPEY SASD+     YV DEWE
Sbjct  58    IMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVNPEYFSASDM-----YVADEWE  222

Query  1020  VPREKITLLRELGQGSFGMVYEGNAKDIIKGEAETRVAVKTVNESASLRERIEFLNEASV  1079
              PRE IT+++ELGQGSFGMVYEG AKDI K E  TRVA+KTVNE A++RERIEFLNEASV
Sbjct  223   FPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVALKTVNELATMRERIEFLNEASV  402

Query  1080  MKGFTCHHVVRLLGVVSKGQPTLVVMELMAHGDLKSHLRSLRPDAENNPGRPPPTLQEMI  1139
             MK F CHHVVRLLGVVS+GQP LV+MELM  GDLKS+LRSLRPD ENN     P+L+ MI
Sbjct  403   MKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLRSLRPDEENNNPGNQPSLKRMI  582

Query  1140  QMTAEIADGMAYLNAKKFVHRDLAARNCMVAHDFTVKIGDFGMTRDIYETDYYRKGGKGL  1199
             QM  EIADGMAYL AKKFVHRDLAARNCMV+ D TVKIGDFGMT DIYETDYYRKGGK L
Sbjct  583   QMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIGDFGMTXDIYETDYYRKGGKSL  762

Query  1200  LPVRWMSPESLKDGVFTASSDMWSFGVVLWEITSLAEQPYQGLSNEQVLKF  1250
             LPVRWMSPESL DG+FT  SD+WSFGVVLWEI +LAEQPYQG++NEQVL F
Sbjct  763   LPVRWMSPESLXDGIFTTHSDVWSFGVVLWEIATLAEQPYQGMANEQVLSF  915
```

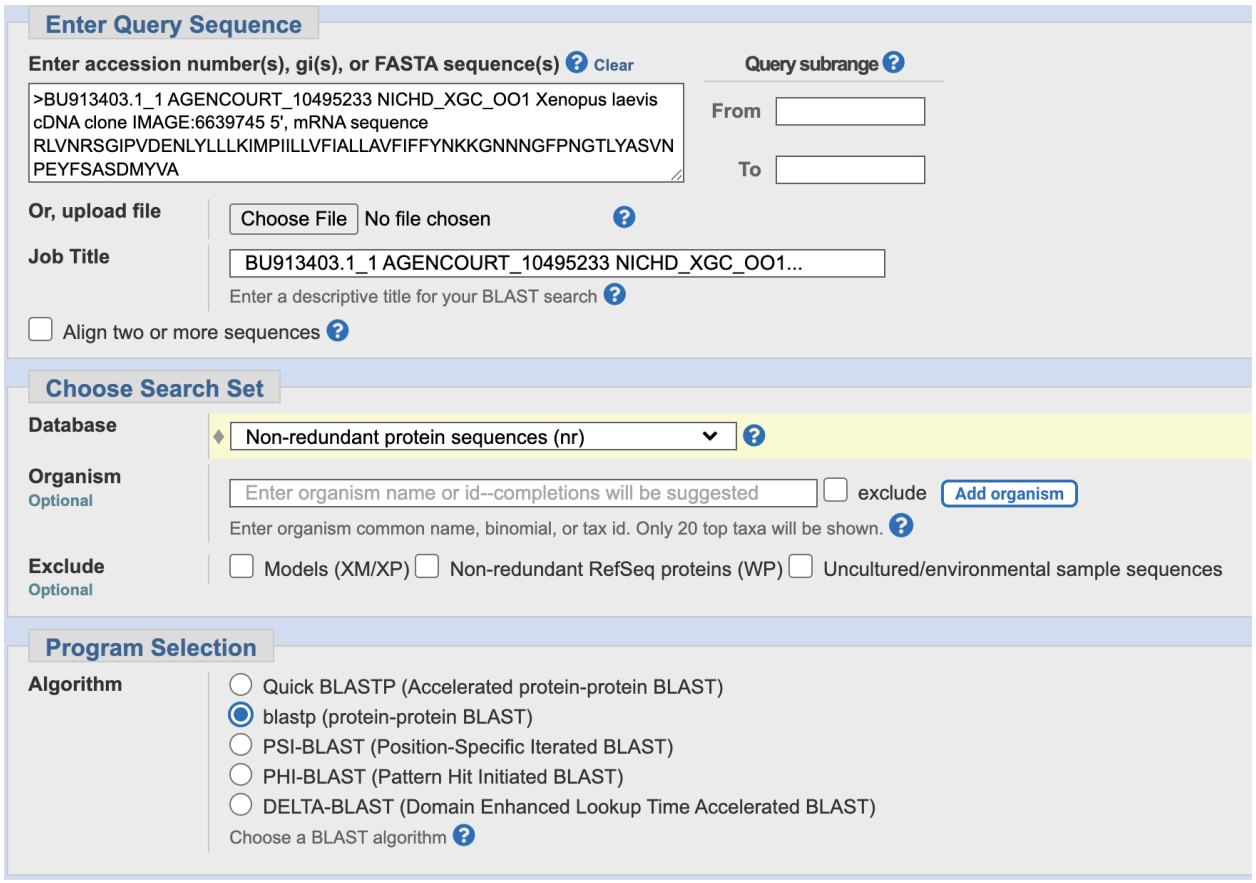
3. From Emboss Transeq (ST), ORF without stop codons:

```
>BU913403.1_1 AGENCOURT_10495233 NICHD_XGC_OO1 Xenopus laevis cDNA clone
IMAGE:6639745 5', mRNA sequence
RLVNRSGIPVDENLYLLLKIMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVNPE
YFSASDMYVADEWEFPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVALKTVNELA
TMRERIEFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLRSLRPDEE
NNNPGNQPSLKRMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIGDFGMTXD
IYETDYYRKGGKSLLPVRWMSPESLXDGIFTTHSDVWSFGVVLWEIATLAEQPYQGMANE
QVLSFCDRITGSWEKPRKLPXIDFMSLMPF
```

Species: Xenopus laevis (African clawed frog)

Protein: Insulin Receptor (unnamed cDNA clone IMAGE:6639745)

4. BLASTP search done with sequence from Q3 at nr database (NCBI)



Results were all LESS than 100% matches, the top result was:
Name: insulin receptor related receptor L homeolog isoform X2 [Xenopus laevis]
Percent identity: 95.79%

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
insulin receptor related receptor L homeolog isoform X2 [Xenopus laevis]	Xenopus laevis	605	605	94%	0.0	95.79%	1345	XP_018084249.1
hypothetical protein XELAEV_18040513mg [Xenopus laevis]	Xenopus laevis	605	605	94%	0.0	95.79%	1358	OCT69203.1
insulin receptor related receptor L homeolog isoform X1 [Xenopus laevis]	Xenopus laevis	605	605	94%	0.0	95.79%	1359	XP_018084247.1
insulin receptor related receptor L homeolog precursor [Xenopus laevis]	Xenopus laevis	605	605	94%	0.0	95.79%	1358	NP_001083465.1
insulin receptor-related protein isoform X3 [Xenopus tropicalis]	Xenopus tropicalis	580	580	94%	0.0	91.59%	1346	XP_004917408.2
hypothetical protein XENTR_v10022447 [Xenopus tropicalis]	Xenopus tropicalis	580	580	94%	0.0	91.59%	1345	KAE8588288.1
insulin receptor-related protein isoform X1 [Xenopus tropicalis]	Xenopus tropicalis	580	580	94%	0.0	91.59%	1360	XP_004917407.2
insulin receptor-related protein isoform X2 [Xenopus tropicalis]	Xenopus tropicalis	580	580	94%	0.0	91.59%	1359	XP_002939040.3
insulin receptor-related protein [Xenopus laevis]	Xenopus laevis	579	579	94%	0.0	91.26%	1359	XP_018088968.1

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insulin receptor related receptor L homeolog isoform X2 [Xenopus laevis]

Sequence ID: XP_018084249.1 Length: 1345 Number of Matches: 1

Range 1: 923 to 1230 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
605 bits(1561)	0.0	Compositional matrix adjust.	296/309(96%)	296/309(95%)	1/309(0%)

```
Query  12    ENLYLLLKIMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVNPEYFSASDMYVAD  71
             ENLYLLLKIMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVNPEYFSASDMYVAD
Sbjct  923   ENLYLLLKIMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVNPEYFSASDMYVAD  982

Query  72    EWEFPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVALKTVNELATMRERIEFLNE  131
             EWEFPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVALKTVNELATMRERIEFLNE
Sbjct  983   EWEFPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVALKTVNELATMRERIEFLNE  1042

Query  132   ASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLRSLRPDEENNNPGNQPSLK  191
             ASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLRSLRPDEENNNPGNQPSLK
Sbjct  1043  ASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLRSLRPDEENNNPGNQPSLK  1102

Query  192   RMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIGDFGMTXDIYETDYYRKGG  251
             RMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIGDFGMT DIYETDYYRKGG
Sbjct  1103  RMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIGDFGMTRDIYETDYYRKGG  1162

Query  252   KSLLPVRWMSPESLXDGIFTTHSDVWSFGVVLWEIATLAEQPYQGMANEQVLSFCDRITG  311
             KSLLPVRWMSPESL DGIFTTHSDVWSFGVVLWEIATLAEQPYQGMANEQVL F     G
Sbjct  1163  KSLLPVRWMSPESLKDGIFTTHSDVWSFGVVLWEIATLAEQPYQGMANEQVLHFVID-NG  1221

Query  312   SWEKPRKLP  320
               EKP   P
Sbjct  1222  ILEKPENCP  1230
```

Related Information

Gene - associated gene details

Genome Data Viewer - aligned genomic context

5) Multiple Alignment Sequence

Re-labeled sequences for alignment:

```
>Query_Mouse_Insulin_Receptor
MGFGRGCETTAVPLLVAVAALLVGTAGHLYPGEVCPGMDIRNNLTRLHELENCSVIEGHLQILLMFKTRP
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VNFSFCQDLHFKCRNSRKPGCHQYVIHNNKCIPECPSGYTMNSSNLMCTPCLGPCPKVCQILEGEKTIDS
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DQASCENELLKFSFIRTSFDKILLRWEPYWPPDFRDLLGFMLFYKEAPYQNVTEFDGQDACGSNSWTVVD
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ISVSNSSSQIILKWKPPSDPNGNITHYLVYWERQAEDSELFELDYCLKGLKLPSRTWSPPFESDDSQKHN
QSEYDDSASECCSCPKTDSQILKELEESSFRKTFEDYLHNVVFVPRKTSSGNGAEDSRPSRKRRSLEEVG
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RGCRLRGLSPGNYSVRVRATSLAGNGSWTEPTYFYVTDYLDVPSNIAKIIIGPLIFVFLFSVVIGSIYLF
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>Novel_Xenopus_laevis
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>ISRR_X3 Xenopus tropicalis | P_004917408.2
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>Rhinophrynus_dorsalis | KAM4690059.1
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>Pelobates_fuscus | XP_063296316.1
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KPPTQRNGNITYYLVLWQQMVEDQELYLNDYCTKGLKLPTSSADTRFDSYDNGNQDANQDSEDKCCPCQK
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YGKMDLK

>Hymenochirus_boettgeri | KAG8449906.1
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QEPSITLTNLKPWTQYAIYVRAITLTTEEKGRNFGAQSDVAYIRTKPAAPTVPQDVMIKSNSSSHLIVRW
KPPTQRNGNITYYLVLWQQMAEDKELYLNDYCTKGLKLPTSSADTRFDNDDNNNQDTNLDTEEKCCPCQK
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DVWSFGVVLWEIATLAEQPYQGMANEQVLHFVIDNGILEKPENCPNRLQELMLWCWQKNPKHRPSFMEIL
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>Engystomops_pustulosus | KAG8543815.1

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KPPWKVTLISKSNQRSPHLLRSRRNRRDVSGIIGETLGNLSLAEGPLSHTINNTAVESKPYTYKDKIFRD
RMVISNLRHFTEYRIDIHACNHAADVVGCSAATFVFARTMPNPTADNIAGNITWEPTGHNTILMKWDEPE
NPNGLILKYEIKYKKEHEEGPSTVVCVSRQKFQKYKGVHLTLVQPGNYSARIRATSLAGNGSWTEPIVFF
VLGPKETPSENIYLLLKVMPILLLLVIACLALFIFFYNKRGKNDGFPSGALYASVNPEYFSASEMYIPDE
WEFPREKITVLKELGQGSFGMVYEGEAKDIVKGEHVTRVALKTVNEMATMRERIEFLNEASVMKAFICHH
VVRLLGVVSQGQPALVIMELMTRGDLKSYLRSLRADDENKVNATPPSLKEMIQMAGEIADGMAYLNAKKF
VHRDLAARNCMVSEDLTVKIGDFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTPYSDVWSFGVV
LWEIATLAEQPYQGMANEQVLHYVMDNGILEKPENCPNRLHELMRWCWQKNPKNRPSFIQILESIKEDLQ
PSFQQLSFFYTTNQKRRESAEVSDAEADQAQKNLLEISPPISSSTSSSSSSFHNFSSLCKAQKDHALLAI
NAKKNPSAKKNIYHMNGSAKH

Comment: *For the following MSA generated by MUSCLE (3.8), the alignment was cropped at the beginning (N-terminus) so that all sequences begin at the same aligned region to improve readability, since the novel sequence begins later in the alignment.

```
Query_Mouse_Insulin_Receptor      TYFYVTDYLDVPSNIAKIIIGPLIFVFLFSVVIGSIYLFLRKRQPDG---PMGPLYASSN
Engystomops_pustulosus            IVFFVLGPKETPSENIYLLLKVMPILLLLVIACLALFIFFYNKRGKNDGFPSGALYASVN
Pelobates_fuscus                  TVFFILGPEEEQSENLYLLLKVMPIVLLLIIALVAVFVFFYNKKGNGEGYPPGTLYASVN
Rhinophrynus_dorsalis             TGFFILGP-DEQSENLYLLLKVMPIILLVIIAFLAVFIFFYNKRGNNDAYPNGTLYASVN
Hymenochirus_boettgeri            VVFFILGPAEEQSENLRLLAIVMPISLLILIVFTALFIFFYNKKGNNNGFPNGTLYASVN
Novel_Xenopus_laevis              -------------ENLYLLLKIMPIILLVFIALLAVFIFFYNKKGNNNGFPNGTLYASVN
ISRR_X3                           MVFFILGPAEEPSENLYLLLKVLPIILLVVIAFLAVFIFFYNKKGNNNGYPNGTLYASVN
                                               :   ::   : : :*. :.  ::::*: :.. ..   * *.**** *

Query_Mouse_Insulin_Receptor      PEYLSASDVFPSSVYVPDEWEVPREKITLLRELGQGSFGMVYEGNAKDIIKGEAETRVAV
Engystomops_pustulosus            PEYFSASE-----MYIPDEWEFPREKITVLKELGQGSFGMVYEGEAKDIVKGEHVTRVAL
Pelobates_fuscus                  PEYLSATD-----MYVADEWEFPREKITIIKELGQGSFGMVYEGVAKDIVKDEPVTKVAL
Rhinophrynus_dorsalis             PEYFSASD-----MYVADEWEFPRENITIIKELGQGSFGMVYEGVAKDIVKGEPVTRVAL
Hymenochirus_boettgeri            PEYLSASD-----MYVPDEWEFPRENITIVKELGQGSFGMVYEGIAKEIVKDEPVTRVAL
Novel_Xenopus_laevis              PEYFSASD-----MYVADEWEFPRENITIMKELGQGSFGMVYEGVAKDIDKNEPTTRVAL
ISRR_X3                           PEYFSASD-----MYVPDEWEFPRENITIIKELGQGSFGMVYEGVAKDIVKNEPVTRVAL
                                  ***:**::     :*:.****.***:**::.************* **:* *.*  *.**:

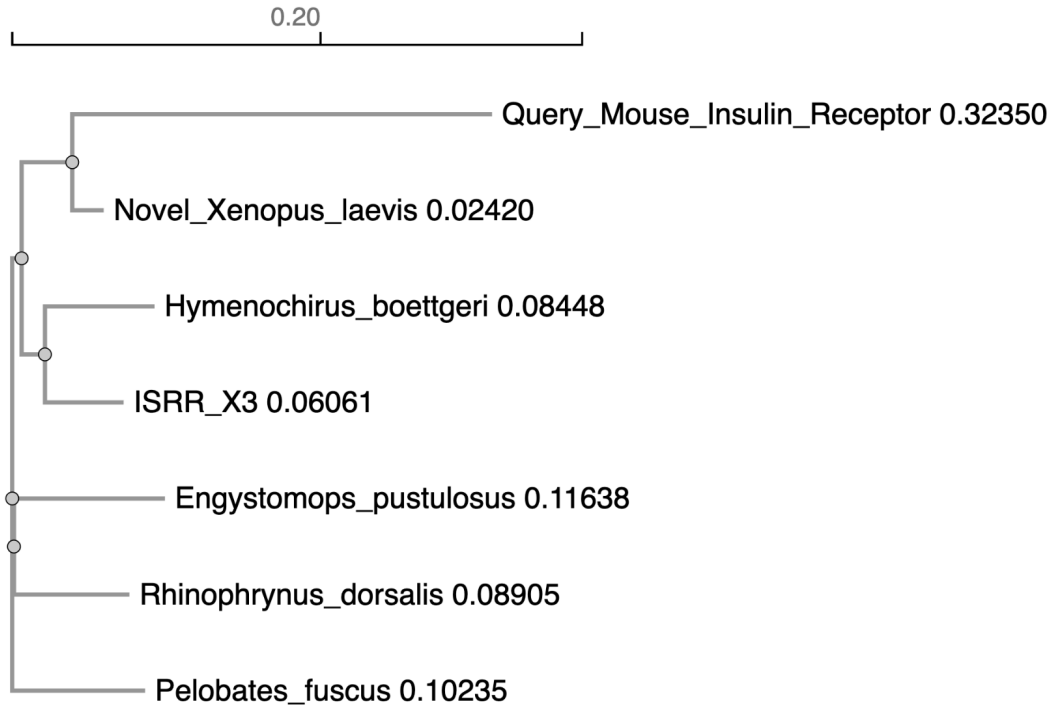
Query_Mouse_Insulin_Receptor      KTVNESASLRERIEFLNEASVMKGFTCHHVVRLLGVVSKGQPTLVVMELMAHGDLKSHLR
Engystomops_pustulosus            KTVNEMATMRERIEFLNEASVMKAFICHHVVRLLGVVSQGQPALVIMELMTRGDLKSYLR
Pelobates_fuscus                  KTVNELASTRERIEFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTRGDLKSYLR
Rhinophrynus_dorsalis             KTVNELASMRERIDFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLR
Hymenochirus_boettgeri            KTVNELATTRERIEFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTRGDLKSYLR
Novel_Xenopus_laevis              KTVNELATMRERIEFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTKGDLKSYLR
ISRR_X3                           KTVNELATTRERIEFLNEASVMKAFLCHHVVRLLGVVSQGQPALVIMELMTRGDLKSYLR
                                  ***** *: ****:*********.* ************:***:**:****:.*****:**

Query_Mouse_Insulin_Receptor      SLRPDAENNPGRPPPTLQEMIQMTAEIADGMAYLNAKKFVHRDLAARNCMVAHDFTVKIG
Engystomops_pustulosus            SLRADDENKVNATPPSLKEMIQMAGEIADGMAYLNAKKFVHRDLAARNCMVSEDLTVKIG
Pelobates_fuscus                  SLRPEDENNSRVSQPSLKRMIQMTGEIADGMAYLNAKKFVHRDLAARNCMVSEDLTVKIG
Rhinophrynus_dorsalis             SLRTEEENNQNASRPTLKEMIQMAGEIADGMAYLNAKKFVHRDLAARNCMVSEDLTVKIG
Hymenochirus_boettgeri            SLRPEEENNFSGAQPPLQQMIQMAGEIADGMAYLNAKKFVHRDLAARNCMVSEDLTVKIG
Novel_Xenopus_laevis              SLRPDEENNNPGNQPSLKRMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIG
ISRR_X3                           SLRPDEENNHPGNQPSLKRMIQMAGEIADGMAYLTAKKFVHRDLAARNCMVSEDLTVKIG
                                  ***.: **:     *.*: ****:.*********.****************: *:*****

Query_Mouse_Insulin_Receptor      DFGMTRDIYETDYYRKGGKGLLPVRWMSPESLKDGVFTASSDMWSFGVVLWEITSLAEQP
Engystomops_pustulosus            DFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTPYSDVWSFGVVLWEIATLAEQP
Pelobates_fuscus                  DFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTLHSDVWSFGVVLWEIATLAEQP
Rhinophrynus_dorsalis             DFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTTHSDVWSFGVVLWEIATLAEQP
Hymenochirus_boettgeri            DFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTTHSDVWSFGVVLWEIATLAEQP
Novel_Xenopus_laevis              DFGMTXDIYETDYYRKGGKSLLPVRWMSPESLXDGIFTTHSDVWSFGVVLWEIATLAEQP
ISRR_X3                           DFGMTRDIYETDYYRKGGKSLLPVRWMSPESLKDGIFTTYSDVWSFGVVLWEIATLAEQP
                                  ***** *************.************ **:**  **:**********::*****

Query_Mouse_Insulin_Receptor      YQGLSNEQVLKFVMDGGYLDPPDNCPERLTDLMRMCWQFNPKMRPT--FLEIVNLLKDDL
Engystomops_pustulosus            YQGMANEQVLHYVMDNGILEKPENCPNRLHELMRWCWQKNPKNRPS--FIQILESIKEDL
Pelobates_fuscus                  YQGLANEQVLHFVIDNGTLEKPDNCPDRLHELMTWCWQKNPKQRPT--FIQILESIKNEL
Rhinophrynus_dorsalis             YQGLANEQVLHFVIDHGILEKPEHCPDRLHELMHWCWQKNPKHRPS--FTQIIESIKEEL
Hymenochirus_boettgeri            YQGMANEQVLHFVIDNGILEKPENCPNRLQELMLWCWQKNPKHRPS--FMEILENIKEDL
Novel_Xenopus_laevis              YQGMANEQVLSFC-------------DRITG----SWEK-PRKLPXIDFMSLMPF-----
ISRR_X3                           YQGMANEQVLHFVIDNGILEKPENCPDRLHELMRWCWQKNPKHRPS--FIQILESIKDEL
                                  ***::***** :              :*:      .*:  *.  *   * .::
```

Q6)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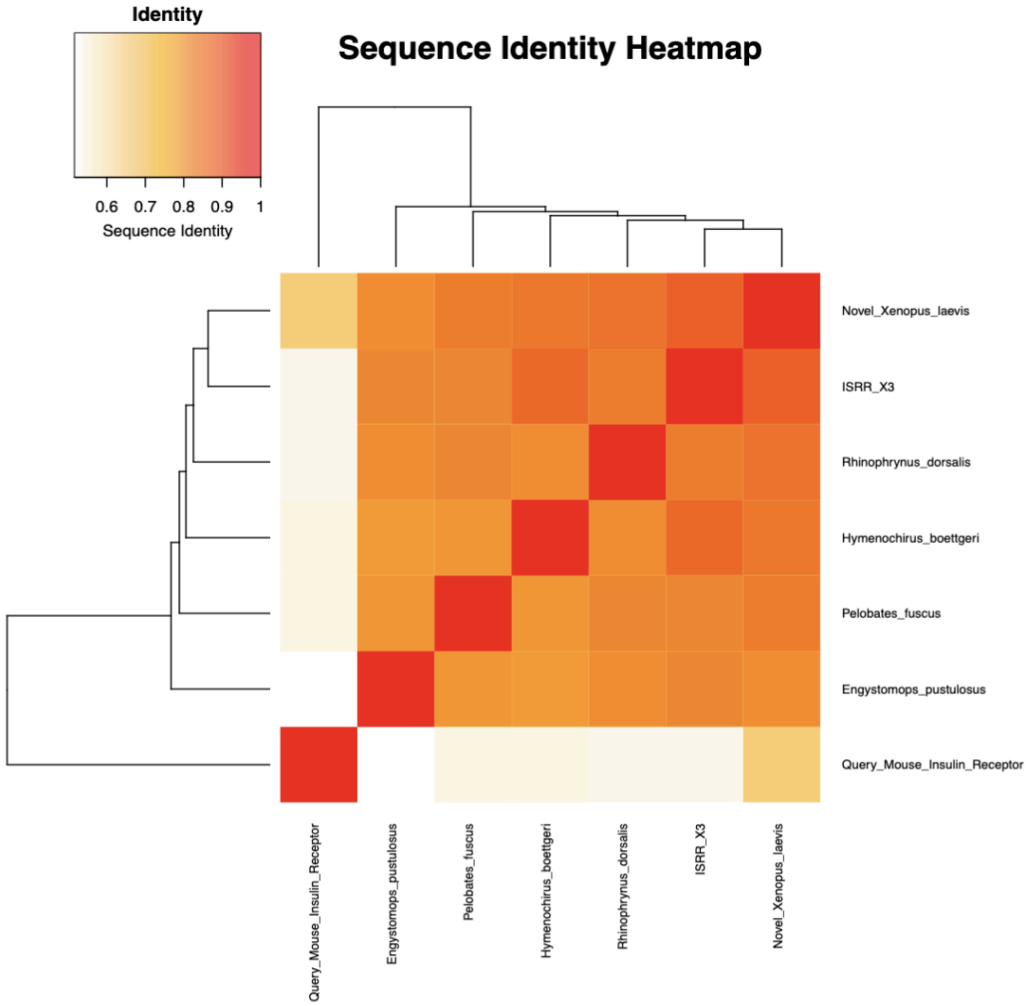
Hymenochirus_boettgeri 0.08448
ISRR_X3 0.06061
Engystomops_pustulosus 0.11638
Rhinophrynus_dorsalis 0.08905
Pelobates_fuscus 0.10235

Q7)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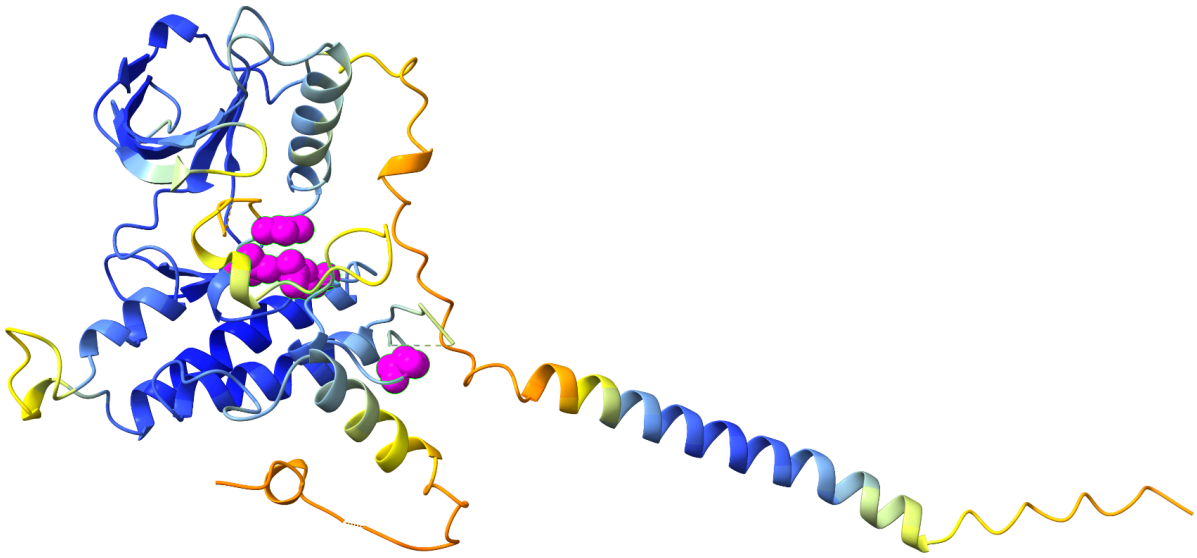
0.6 0.7 0.8 0.9 1
Sequence Identity
Sequence Identity Heatmap
Novel_Xenopus_laevis
ISRR_X3
Rhinophrynus_dorsalis
Hymenochirus_boettgeri
Pelobates_fuscus
Engystomops_pustulosus
Query_Mouse_Insulin_Receptor
Query_Mouse_Insulin_Receptor
Engystomops_pustulosus
Pelobates_fuscus
Hymenochirus_boettgeri
Rhinophrynus_dorsalis
ISRR_X3
Novel_Xenopus_laevis

Q8)

ID	E-value	Identity	Source	Technique	Resolution
3LVP	4.58e-162	75.768	Homo sapiens	X-ray Diffraction	3 Å
8PYI	1.57e-154	79.259	Homo sapiens	X-ray Diffraction	3 Å
1M7N	2.21e-154	79.259	Homo sapiens	X-ray Diffraction	3 Å

Q9) Generated Structural Model using AlphaFold Notebook and ChimeraX. Conserved residues are shown in magenta spacefill and the figure is colored by local alpha fold pLDDT quality score.

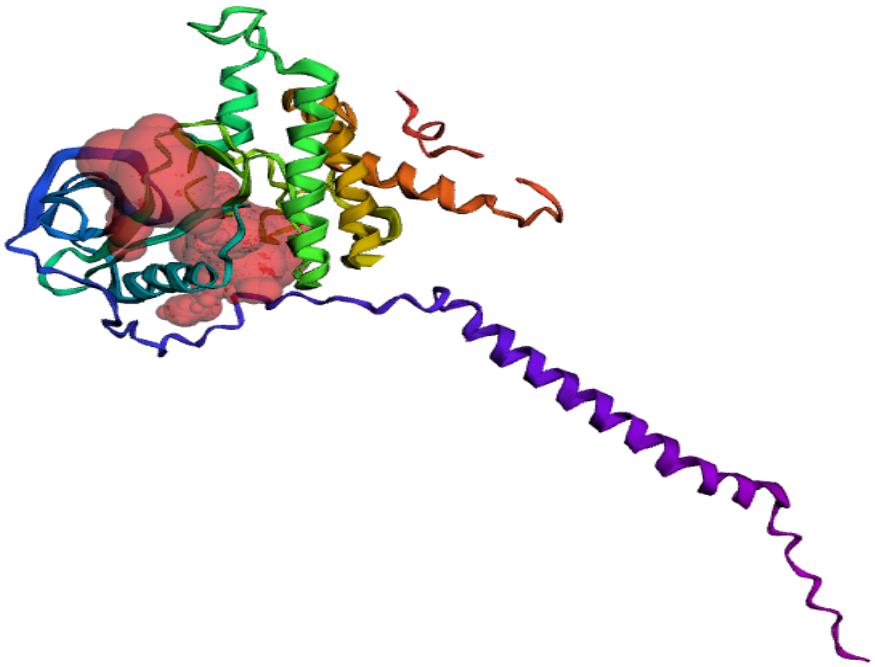


Q10 (i). Below are the largest predicted pockets

	Pocket ID	Area (SA) (Å^2)	Volume (SA) (Å^3)
+	1	1149.062	1241.854
+	2	547.986	725.027
+	3	275.404	142.182
+	4	180.314	81.933
+	5	105.250	45.541

Based on the table:

Pocket 1 with Area (SA) = 1149.062 and Volume (SA) = 1241.854 is highlighted in red in the structure:



10(ii). Non available as of March 6, 2026.

10(iii). The presence of well-defined pockets identified by CASTpFold with a relatively large surface area and volume suggests potential ligand binding capability. Conservation analysis indicates that several residues within this pocket are conserved, suggesting that this region may play an important functional role in the protein. A search of the ChEMBL database did not reveal any associated assays or inhibitors for this protein, indicating that it may represent a novel or uncharacterized target that has not been studied as a drug target. However, the presence of a defined pocket and conserved residues suggests potential druggability and possible relevance for therapeutic targeting.